

NESA Number: _____ Name: _____

Teacher: MA MN RC LD



Ascham School

Mathematics Extension 1

Trial HSC Examination

Monday 28th July 2025

2 hours

General Instructions

- Reading time – 10 minutes.
- Working time – 2 hours.
- Write using black non-erasable pen.
- NESA-approved calculators may be used.
- A Reference Sheet is provided.
- In Questions 11–14, show relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I Pages 3 – 7

10 marks

- Use the Multiple-Choice Answer Sheet provided to answer Q1-10.
- Allow about 15 minutes for this section.

Section II Pages 8 – 13

60 marks

- Answer Questions 11-14.
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher's initials.
- Allow about 1 hour and 45 minutes for this section.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which is the cartesian equation representing the following parametric equations?

$$\begin{aligned}x &= 3t - 1 \\ y &= \frac{1}{t} \quad \text{where } t \neq 0\end{aligned}$$

(A) $y = \frac{3}{x} + 1$

(B) $y = \frac{3}{x} - 1$

(C) $y = \frac{3}{x + 1}$

(D) $y = \frac{x + 1}{3}$

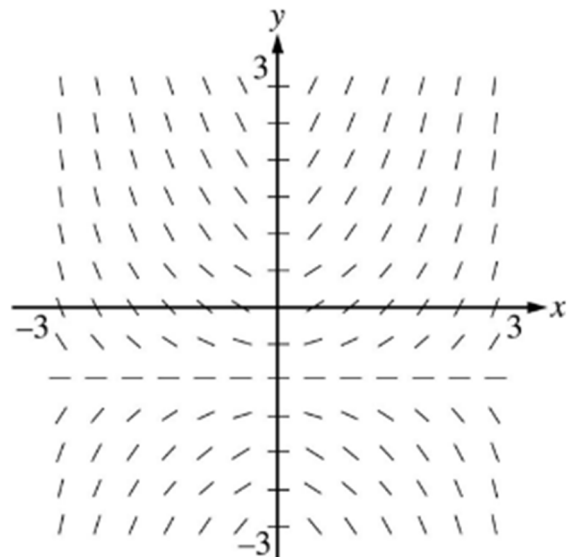
- 2 Which differential equation represents the slope field graphed?

(A) $\frac{dy}{dx} = (x + 1)^2$

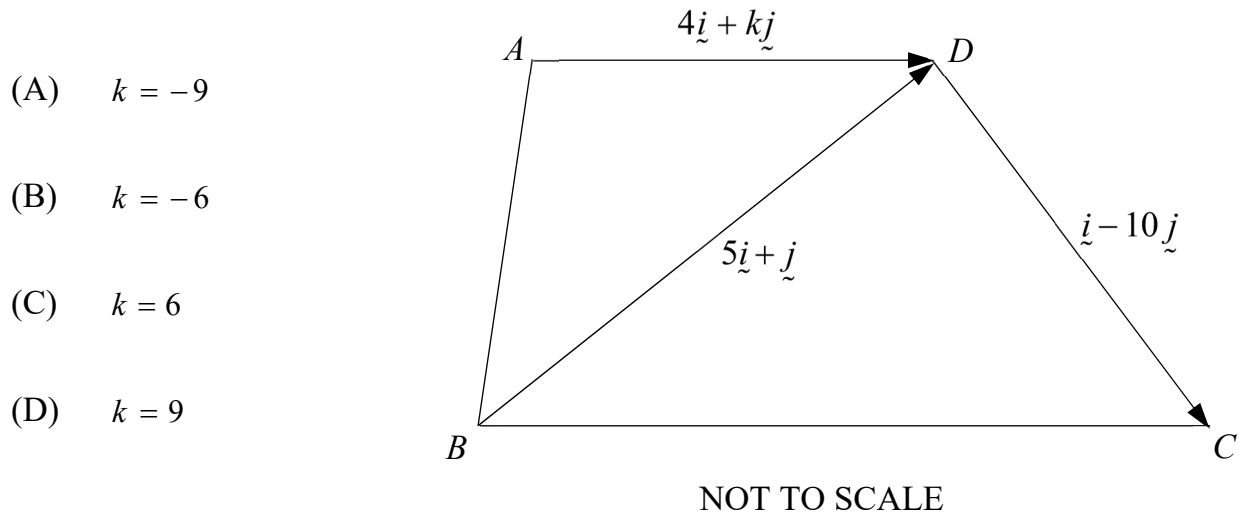
(B) $\frac{dy}{dx} = y + 1$

(C) $\frac{dy}{dx} = xy + y$

(D) $\frac{dy}{dx} = xy + x$



- 3 The figure shows a trapezium $ABCD$, where $\underline{AD}$ is parallel to $\underline{BC}$.
 $\overrightarrow{BD} = 5\mathbf{i} + \mathbf{j}$, $\overrightarrow{DC} = \mathbf{i} - 10\mathbf{j}$ and $\overrightarrow{AD} = 4\mathbf{i} + k\mathbf{j}$, where k is an integer.
 What is the correct value of k ?

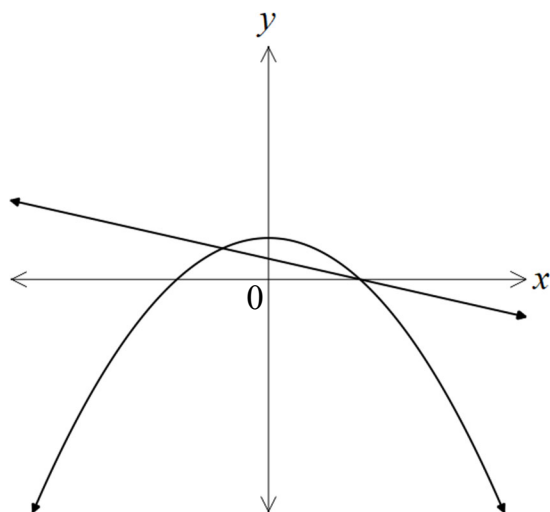


- 4 Which of the following expressions is equivalent to $2\sqrt{3}\cos x - 2\sin x$?

- (A) $4\cos\left(x + \frac{\pi}{6}\right)$
- (B) $4\sqrt{3}\cos\left(x + \frac{\pi}{6}\right)$
- (C) $4\cos\left(x + \frac{\pi}{3}\right)$
- (D) $4\sqrt{3}\cos\left(x + \frac{\pi}{3}\right)$

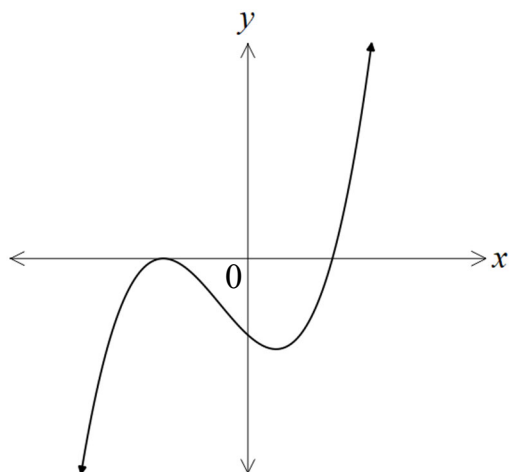
- 5 Two adults and six children are seated at a circular dining table.
In how many ways can they be arranged so that the adults are not seated next to each other?
- (A) 2880
- (B) 3600
- (C) 4320
- (D) 25 200
- 6 An anti-derivative of $\frac{1}{x^2 - 4x + 13}$ is:
- (A) $\tan^{-1}\left(\frac{x-2}{3}\right)$
- (B) $\tan^{-1}\left(\frac{x-3}{2}\right)$
- (C) $\frac{1}{2}\tan^{-1}\left(\frac{x-3}{2}\right)$
- (D) $\frac{1}{3}\tan^{-1}\left(\frac{x-2}{3}\right)$
- 7 Suppose $x^3 - 4x^2 + a \equiv (x+1)Q(x) - 2$, where $Q(x)$ is a polynomial.
The correct value of a is:
- (A) -3
- (B) 1
- (C) 3
- (D) 5

- 8 The graphs of $y = f(x)$ and $y = g(x)$ are shown below.

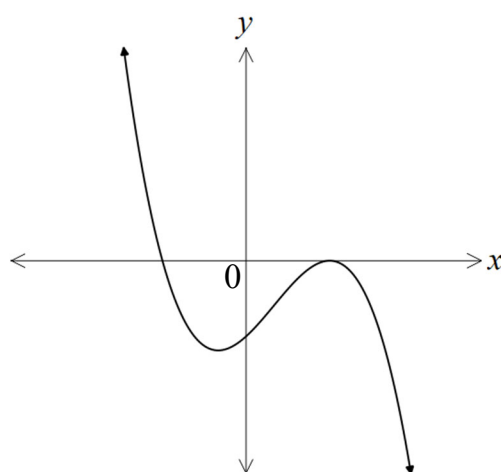


Which of the following graphs could present $y = f(x)g(x)$?

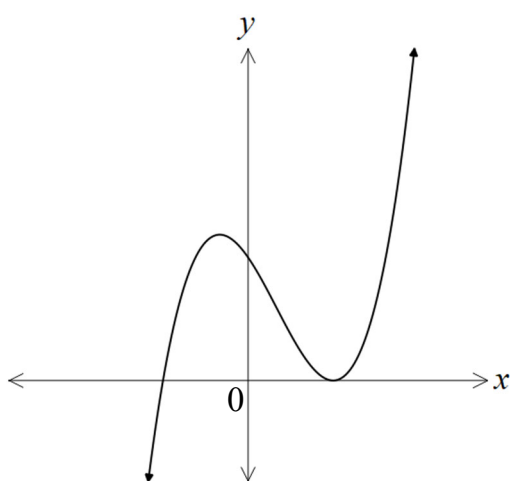
(A)



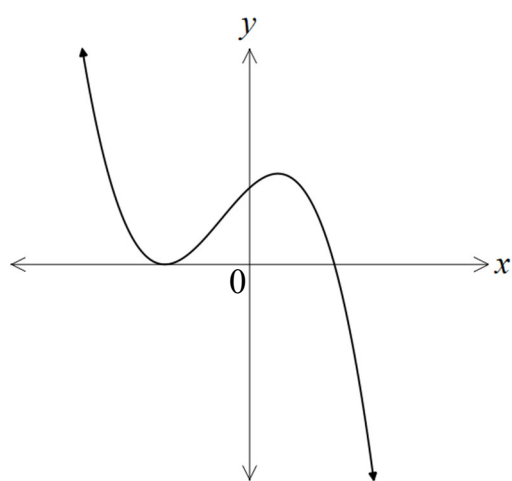
(B)



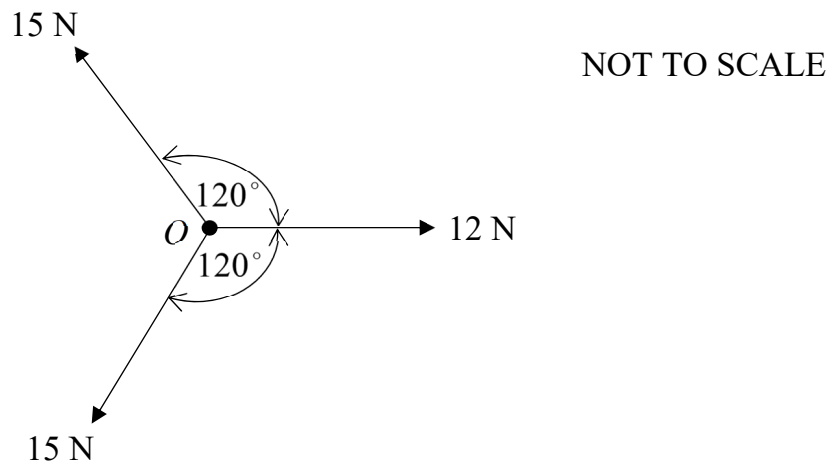
(C)



(D)



- 9 A bag contains 3 red marbles, 4 green marbles, 6 pink marbles and 7 black marbles. Marbles are drawn at random one at a time without replacement from the bag. What is the least number of marbles that need to be drawn to be certain that 6 marbles of the same colour are amongst those selected?
- (A) 17
- (B) 18
- (C) 19
- (D) 21
- 10 The figure below shows three forces which lie on the same plane, acting on a particle at O . The magnitude of these forces and their relative directions are shown.



What is the magnitude of the resultant of the three forces?

- (A) 3 N
- (B) 9 N
- (C) 12 N
- (D) 27 N

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.
In Questions 11–14, you should include relevant mathematical reasoning and/ or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Find:

(i) $\int \frac{x}{\sqrt{1-x^4}} dx.$ 2

(ii) $\int 12 \sin 4x \sin 2x dx.$ 2

(b) Solve $\frac{1-2x}{x-5} \geq 1.$ 3

(c) (i) Write down the expansion of $(x-2)^5.$ 2

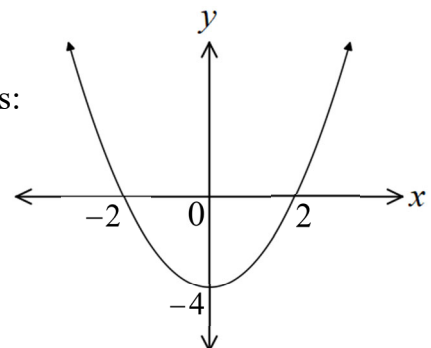
(ii) Hence or otherwise, find the coefficient of x in the expansion of $(2x+1)^2(x-2)^5.$ 1

(d) The polynomial $P(x) = 10x^3 + ax^2 - 20x + b$ has $x = 2$ as a root of multiplicity 2.
Find the values of a and $b.$ 2

(e) Consider the graph of the function $f(x) = x^2 - 4.$
Sketch the following functions showing all key features:

(i) $y = |f(x)|$ 1

(ii) $y = \frac{1}{|f(x)|}$ 2



Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) 8 females and 3 males are members of a mixed netball training squad. However, only 7 players make up a starting team. How many ways can a starting team be formed if there must be at least 2 males playing? 2

- (b) Consider the function $y = 2 \cos^{-1}(2x-1)$.

(i) Find its domain and range. 2

(ii) Sketch the graph of $y = f(x)$. 1

(iii) How many solutions are there for the equation $2 \cos^{-1}(2x-1) = -2\pi x + 2\pi$? 1

- (c) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, solve the equation:

$$5 \sin x - 12 \cos x = 12, \quad \text{for } 0^\circ \leq x \leq 360^\circ.$$

Give your answers correct to the nearest minute. 3

- (d) Consider the position vector $\overrightarrow{OP} = 3\hat{i} + 4\hat{j}$ and the line l which has equation $x - 2y = 0$.

(i) Find the projection of the vector $\overrightarrow{OP}$ onto the line l . 2

(ii) Hence find the exact perpendicular distance from point P to the line l . 1

- (e) The gradient at any point (x, y) on a curve $y = f(x)$ is given as:

$$\frac{dy}{dx} = k\sqrt{y}, \quad \text{where } k \text{ is a constant.}$$

Find the equation of $y = f(x)$ if it passes through the points $P(4, 4)$ and $Q(6, 16)$. 3

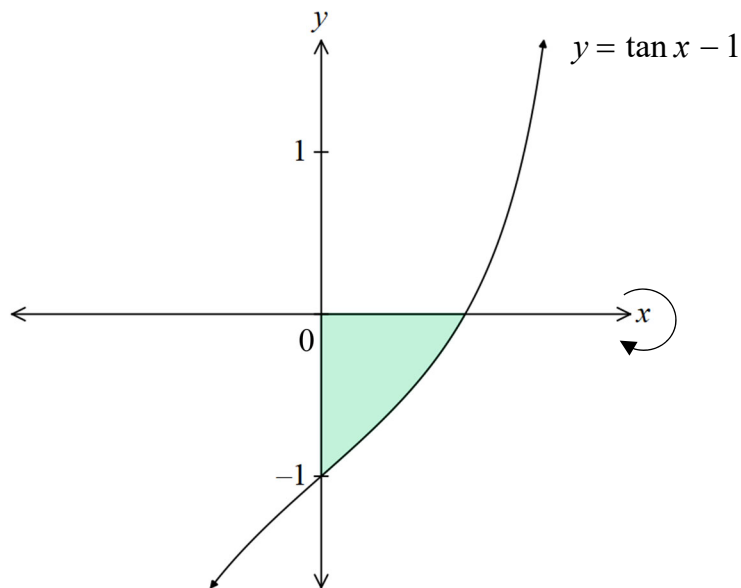
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) A curve has equation $x = y\sqrt{2-y}$, where $y \leq 2$. Find $\frac{dy}{dx}$ at the point $(-4, -2)$. **2**

- (b) Prove by mathematical induction that for all integers $n \geq 1$:

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n} \quad \textbf{3}$$

- (c) The region bounded by the curve $y = \tan x - 1$, the x -axis and the y -axis is shaded below. Find the exact volume of the solid formed if this region is rotated about the x -axis. **3**



- (d) Liquid is pouring into a container at the constant rate of $12\pi \text{ cm}^3/\text{s}$. **3**
The container is initially empty and when the height of the liquid in the container is h cm, the volume of the liquid, $V \text{ cm}^3$, is given by:

$$V = \pi h(h + 20).$$

Determine the rate at which the height of the liquid in the container is rising 8 seconds after the liquid started pouring in.

Question 13 continues on the following page.

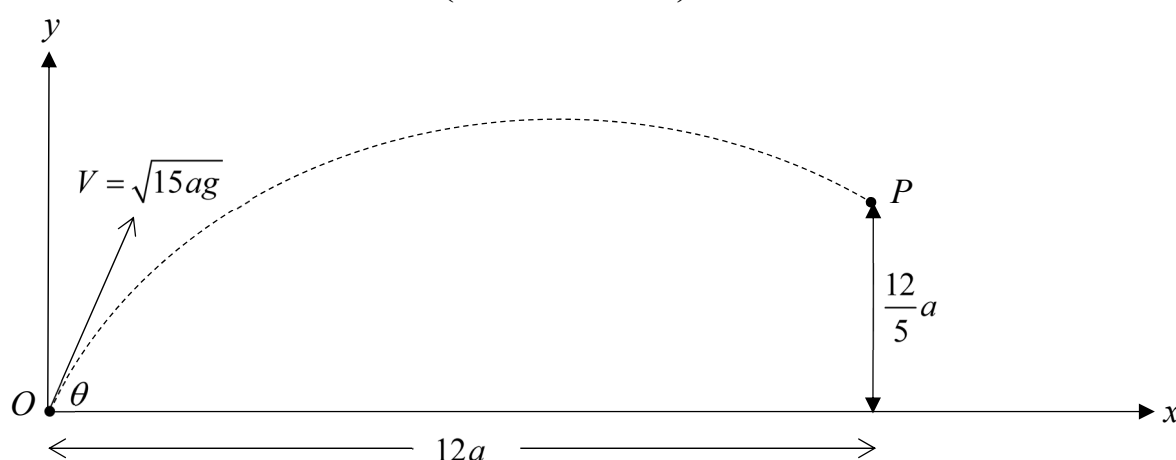
- (e) A projectile is fired from a fixed point O at an angle of elevation θ .

It passes through the point P where $\vec{OP} = \begin{pmatrix} 12a \\ \frac{12}{5}a \end{pmatrix}$.

The speed of its projection is $V = \sqrt{15ag}$, where $a > 0$ and $g \text{ m/s}^2$ is the acceleration due to gravity.

The position vector of the projectile is given below. It shows the horizontal and vertical displacements, x m and y m from O at time t seconds.

$$\underline{s}(t) = \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}$$



Find the possible values of θ correct to the nearest degree.

4

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^1 \sqrt{\frac{x}{1-x}} dx$ using the substitution $x = \sin^2 \theta$. 4

(Assume that $0 \leq \sin \theta \leq 1$).

(b) Solve the following equation for x :

$$\tan^{-1}\left(\frac{1}{x}\right) + \tan^{-1}\left(\frac{1}{x+1}\right) = \frac{\pi}{4} \quad 3$$

(c) The population P of a colony of birds, in thousands, varies according to the differential equation:

$$\frac{dP}{dt} = Pe^{-0.5t}$$

where $P > 0$, $t \geq 0$ and t is the time in years.

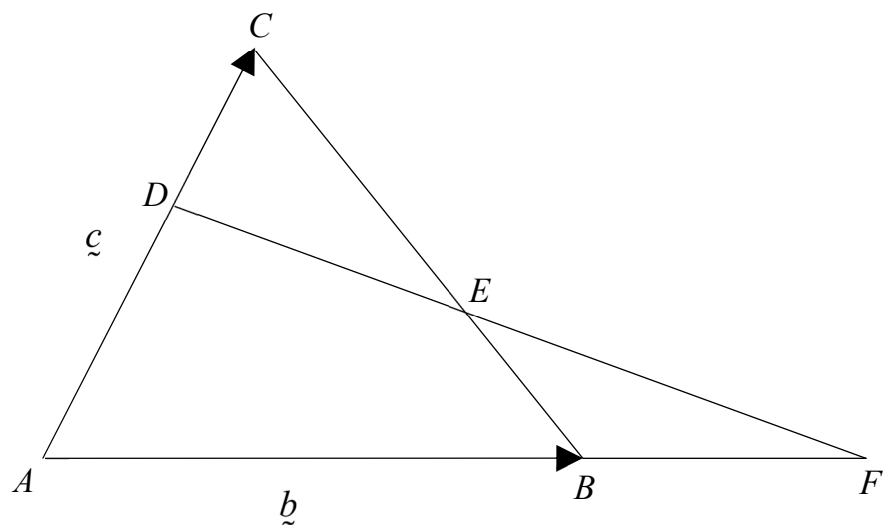
(i) Using separation of variables, solve the given differential equation to show that:

$$P = Ae^{-2e^{-0.5t}} \quad \text{where } A \text{ is a constant.} \quad 2$$

(ii) P approaches its limiting value as $t \rightarrow \infty$.
Find the time when P will reach half its limiting value.
Give your answer in exact form. 2

Question 14 continues on the following page.

(d)



In $\triangle ABC$ above, D is the point on AC such that $AD : DC = 2 : 1$ and E is the point on BC such that $BE : EC = 1 : 2$.

When DE is extended, it meets the extension of AB at F .

Let $\overrightarrow{AB} = \underset{\sim}{b}$ and $\overrightarrow{AC} = \underset{\sim}{c}$.

(i) Show that $\overrightarrow{DE} = \frac{2}{3}\underset{\sim}{b} - \frac{1}{3}\underset{\sim}{c}$. **1**

(ii) By considering the collinearity of DEF and ABF , show that $AB : BF = 3 : 1$. **3**

End of Exam

Year 12 Ext. 1 Trial Exam 2025 SOLUTIONS

Section I

1 If $y = \frac{1}{t}$, then $t = \frac{1}{y}$

and if $x = 3t - 1$

then $x = 3\left(\frac{1}{y}\right) - 1$

$$x + 1 = \frac{3}{y}$$

$$y = \frac{3}{x+1} \quad \therefore \underline{\underline{C}}$$

2 D

4

Let $2\sqrt{3}\cos x - 2\sin x = R\cos(x+\alpha)$

$$2\sqrt{3}\cos x - 2\sin x = R[\cos x \cos \alpha - \sin x \sin \alpha]$$

$\therefore$ Equating Co-efficients:

$$R\cos \alpha = 2\sqrt{3} \quad \dots (1)$$

$$R\sin \alpha = 2 \quad \dots (2)$$

$$(1)^2 + (2)^2 \Rightarrow R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = (2\sqrt{3})^2 + 2^2$$

$$R^2(1) = 4 \times 3 + 4$$

$$R^2 = 16$$

$$\therefore R = 4$$

$$(2) \div (1): \frac{R\sin \alpha}{R\cos \alpha} = \frac{2}{2\sqrt{3}}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\text{i.e. } 2\sqrt{3}\cos x - 2\sin x = 4\cos\left(x + \frac{\pi}{6}\right)$$

$$\therefore \underline{\underline{A}}$$

3 $\vec{BC} = \begin{pmatrix} 5 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -10 \end{pmatrix}$
 $= \begin{pmatrix} 6 \\ -9 \end{pmatrix}$

If $\vec{BC} \parallel \vec{AD}$, then

$$\begin{pmatrix} 4 \\ k \end{pmatrix} = \lambda \begin{pmatrix} 6 \\ -9 \end{pmatrix}$$

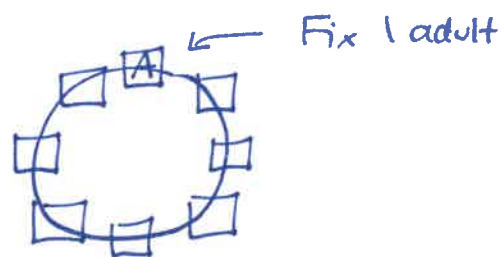
or gradients being parallel means

$$\frac{k}{4} = \frac{-9}{6}$$

$$k = -\frac{9}{6} \times 4$$

$$\therefore k = -6 \quad \therefore \underline{\underline{B}}$$

5



2nd adult can be seated in 5 spots

Then remaining 6 children can be arranged in $6!$ ways

$$5 \times 6!$$

$$= 3600 \text{ ways}$$

$$\therefore \underline{\underline{B}}$$

6

$$\begin{aligned} & \frac{1}{x^2 - 4x + 13} \\ &= \frac{1}{x^2 - 4x + 4 + 13 - 4} \\ &= \frac{1}{(x-2)^2 + 9} \end{aligned}$$

$$\therefore \int \frac{1}{x^2 - 4x + 13} dx$$

$$= \int \frac{1}{(x-2)^2 + 3^2} dx$$

$$= \frac{1}{3} \tan^{-1} \left(\frac{x-2}{3} \right)$$

$\therefore \underline{\underline{D}}$

7 Substitute $x = -1$ into the identity:

$$x^3 - 4x^2 + a \equiv (x+1)Q(x) - 2$$

$$\text{i.e. } (-1)^3 - 4(-1)^2 + a \equiv (-1+1)Q(-1) - 2$$

$$-1 - 4 + a \equiv 0 - 2$$

$$-5 + a \equiv -2$$

$$\therefore a = 3$$

$\therefore \underline{\underline{C}}$

8

C

9

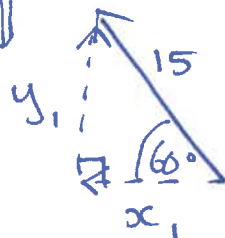
Least number of marbles to ensure 6 of same colour

$$= 3 \text{ red} + 4 \text{ green} + 5 \text{ pink} + 5 \text{ black} + 1 \text{ (pink or black)}$$

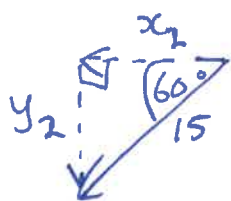
$$= 18$$

$\therefore \underline{\underline{B}}$

10



$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} -15 \cos 60^\circ \\ 15 \sin 60^\circ \end{pmatrix}$$



$$\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} -15 \cos 60^\circ \\ -15 \sin 60^\circ \end{pmatrix}$$

$$\xrightarrow{\frac{12}{x_3}} \begin{pmatrix} x_3 \\ y_3 \end{pmatrix} = \begin{pmatrix} 12 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \end{pmatrix} = \begin{pmatrix} -15 \cos 60^\circ - 15 \cos 60^\circ + 12 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -30 \cos 60^\circ + 12 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -30 \left(\frac{1}{2} \right) + 12 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ 0 \end{pmatrix}$$

$\therefore \text{Magnitude} = 3$

$\therefore \underline{\underline{A}}$



Ascham School

NESA Number: _____

Name: SOLUTIONS

Teacher: MA MN RC LD

SECTION I

Year 12 Extension 1 Mathematics Multiple Choice Answer Sheet

2025 TRIAL EXAMINATION

- | | | | | |
|----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 | A ☐ | B ☐ | C ☒ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☒ |
| 3 | A ☐ | B ☒ | C ☐ | D ☐ |
| 4 | A ☒ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☒ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☒ |
| 7 | A ☐ | B ☐ | C ☒ | D ☐ |
| 8 | A ☐ | B ☐ | C ☒ | D ☐ |
| 9 | A ☐ | B ☒ | C ☐ | D ☐ |
| 10 | A ☒ | B ☐ | C ☐ | D ☐ |

Section II

Question 11

a) i) $\int \frac{x}{\sqrt{1-x^4}} dx$

$$= \int \frac{x}{\sqrt{1-(x^2)^2}} dx$$

$$= \frac{1}{2} \int \frac{2x}{\sqrt{1-(x^2)^2}} dx$$

$$= \frac{1}{2} \sin^{-1}(x^2) + C$$

(2 marks)

ii) $\int 12 \sin 4x \sin 2x dx$

$$= 12 \int \frac{1}{2} [\cos(4x-2x) - \cos(4x+2x)] dx$$

$$= 6 \int (\cos 2x - \cos 6x) dx$$

$$= 6 \left[\frac{\sin 2x}{2} - \frac{\sin 6x}{6} \right] + C$$

$$= 3 \sin 2x - \sin 6x + C$$

(2 marks)

c) i) $(x-2)^5 = \binom{5}{0} x^5 (-2)^0 + \binom{5}{1} x^4 (-2)^1 + \binom{5}{2} x^3 (-2)^2 + \binom{5}{3} x^2 (-2)^3$
 $+ \binom{5}{4} x (-2)^4 + \binom{5}{5} (-2)^5$

$$= x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$$

(2 marks)

ii) $(2x+1)^2 (x-2)^5$

$$= (4x^2 + 4x + 1)(x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32)$$

Term in x is: $4x(-32) + 1(80x)$

$$= -48x$$

$\therefore$ coefficient is -48

(1 mark)

b) $\frac{1-2x}{x-5} \geq 1 \quad x \neq 5$

$$\frac{(1-2x)(x-5)^2}{x-5} \geq (x-5)^2$$

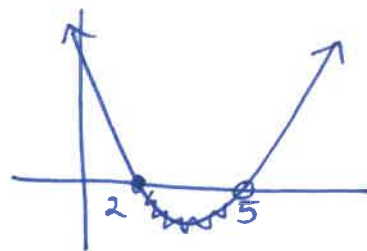
$$(1-2x)(x-5)^2 \geq (x-5)^2$$

$$(x-5)^2 - (1-2x)(x-5) \leq 0$$

$$(x-5)[x-5-(1-2x)] \leq 0$$

$$(x-5)(3x-6) \leq 0$$

$$3(x-5)(x-2) \leq 0$$



$$\therefore 2 \leq x \leq 5$$

(3 marks)

d) If $P(x)$ has $x=2$ as a root of multiplicity 2, then

$$P(2) = 0 \text{ and } P'(2) = 0$$

$$P(2) = 10(2)^3 + a(2)^2 - 20(2) + b$$

$$= 80 + 4a - 40 + b$$

$$\therefore P(2) = 4a + b + 40$$

$$P'(x) = 30x^2 + 2ax - 20$$

$$\therefore P'(2) = 30(2)^2 + 2a(2) - 20$$

$$= 120 + 4a - 20$$

$$\therefore P'(2) = 100 + 4a$$

Now if $P(2) = 0$, then

$$4a + b + 40 = 0$$

$$4a + b = -40 \dots (1)$$

And if $P'(2) = 0$, then

$$100 + 4a = 0 \dots (2)$$

$$\therefore \underline{a = -25}$$

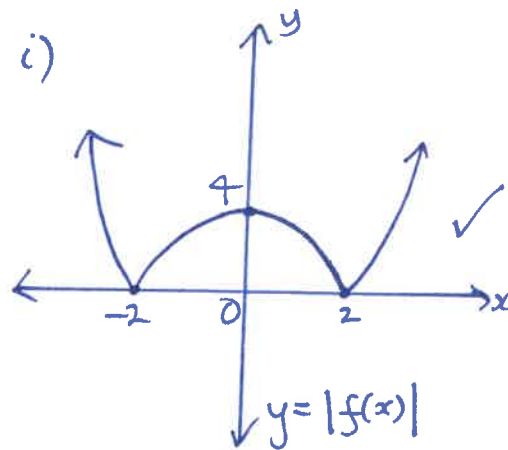
Sub. into (1):

$$4(-25) + b = -40$$

$$b = -40 + 100$$

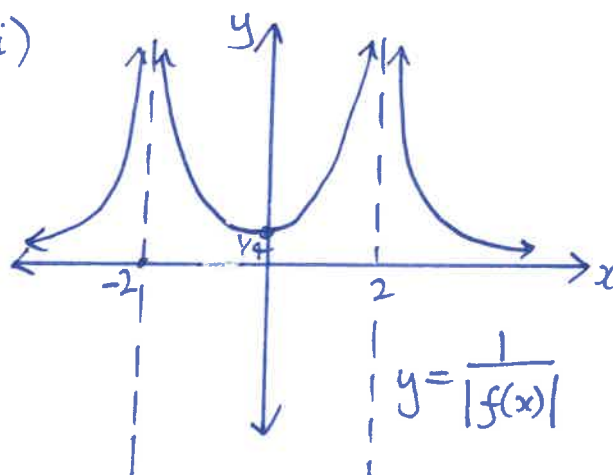
$$\therefore \underline{b = 60}$$

e) i)



(1 mark)

ii)



✓ shape
✓ asymptotes/intercepts

(2 marks)

(2 marks)

Question 12

- a) Either 2 males / 5 females
or 3 males / 4 females

$$\text{No. ways} = {}^3C_2 \times {}^8C_5 + {}^3C_3 \times {}^8C_4$$

$$= 3 \times 56 + 1 \times 70$$

$$= 238 \text{ ways}$$

✓ (at least 1 pair)

✓ (correct answer)

(2 marks)

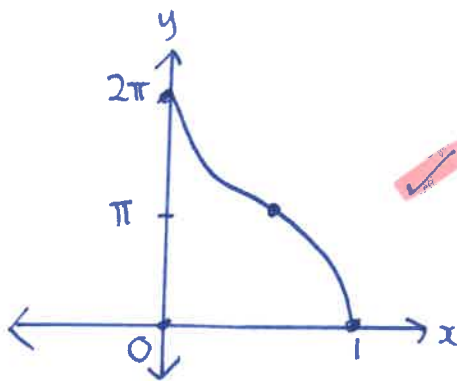
b) i) Domain: $-1 \leq 2x-1 \leq 1$
 $0 \leq 2x \leq 2$

$$\therefore 0 \leq x \leq 1$$

Range: $0 \leq y \leq 2\pi$

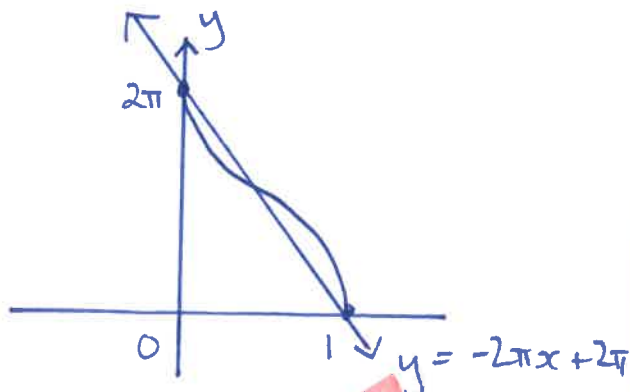
(2 marks)

ii)



(1 mark)

iii)



$\therefore$ 3 solutions
due to 3 points of intersection

(1 mark)

$$c) 5 \sin x - 12 \cos x = 12$$

$$\Rightarrow 5\left(\frac{2t}{1+t^2}\right) - 12\left(\frac{1-t^2}{1+t^2}\right) = 12$$

$$\left[\text{when } t = \tan \frac{x}{2} \right]$$

$$10t - 12(1-t^2) = 12(1+t^2)$$

$$10t - 12 + 12t^2 = 12 + 12t^2$$

$$10t - 12 - 12 = 0$$

$$10t = 24 \leftarrow \text{linear}$$

$$\therefore \text{test } x = 180^\circ$$

$$\therefore t = \frac{12}{5}$$

$$\therefore \tan\left(\frac{x}{2}\right) = \frac{12}{5}$$

where if $0^\circ \leq x \leq 360^\circ$

then $0^\circ \leq \frac{x}{2} \leq 180^\circ$

$$\therefore \frac{x}{2} = 67.3...^\circ \text{ (1st quad only)}$$

$$x = 67.3... \times 2$$

$$x = 134^\circ 45' 36.97''$$

$$\therefore \underline{x \approx 134^\circ 46'}$$

(nearest minute)

Also need to test $x = 180^\circ$:

$$\text{LHS} = 5 \sin(180^\circ) - 12 \cos(180^\circ)$$

$$= 5(0) - 12(-1)$$

$$= 12$$

$$= \text{RHS}$$

$$\therefore \underline{x = 180^\circ} \text{ is also soln}$$

i.e. 2 solutions are

$$x = 180^\circ \text{ or } 134^\circ 46'$$

(3 marks)

d) The line $x - 2y = 0$
 $\Rightarrow 2y = x$
 $\Rightarrow y = \frac{1}{2}x$

has gradient $= \frac{1}{2}$

$(\frac{1}{2})$

$\therefore$ vector parallel to this line is $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

i) projection vector $= \frac{\begin{pmatrix} 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}}{\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

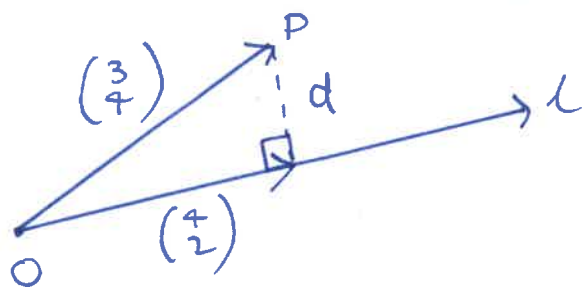
$= \frac{6+4}{4+1} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$= 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ or $4\mathbf{i} + 2\mathbf{j}$ $(\frac{1}{2})$

(2 marks)

ii)



$|\begin{pmatrix} 3 \\ 4 \end{pmatrix}| = \sqrt{3^2 + 4^2}$
 $= \sqrt{25}$
 $= 5$

$|\begin{pmatrix} 4 \\ 2 \end{pmatrix}| = \sqrt{4^2 + 2^2}$
 $= \sqrt{20}$

$d^2 + (\sqrt{20})^2 = 5^2$

$d^2 + 20 = 25$

$d^2 = 5$

$\therefore d = \sqrt{5}$ units

(1 mark)

e) $\frac{dy}{dx} = k\sqrt{y}$

$\frac{1}{\sqrt{y}} dy = k dx$

$\int y^{-1/2} dy = \int k dx$

$\frac{y^{1/2}}{1/2} = kx + c$

$2\sqrt{y} = kx + c$

• when $x=4, y=4$

$\therefore 2\sqrt{4} = 4k + c$

$4k + c = 4 \dots (1)$

• when $x=6, y=16$

$\therefore 2\sqrt{16} = 6k + c$

$6k + c = 8 \dots (2)$

$(2) - (1): 2k = 4$

$\therefore k = 2$

$\Rightarrow (1): 4(2) + c = 4$

$\therefore c = -4$

$\therefore 2\sqrt{y} = 2x - 4$

$\sqrt{y} = \frac{2(x-2)}{2}$

$\sqrt{y} = x - 2$

$\therefore y = (x-2)^2$

$x \geq 2$

$x \geq 2$

(3 marks)

Question 13

a) If $x = y(2-y)^{\frac{1}{2}}$

$$\frac{dx}{dy} = u'v + v'u$$

where $u = y$ $v = (2-y)^{\frac{1}{2}}$
 $\therefore u' = 1$ $v' = \frac{1}{2}(2-y)^{-\frac{1}{2}} \times -1$

$$\therefore \frac{dx}{dy} = \sqrt{2-y} - \frac{y}{2\sqrt{2-y}} \quad \checkmark$$

when $y = -2$:

$$\frac{dx}{dy} = \sqrt{2-(-2)} - \frac{-2}{2\sqrt{2-(-2)}}$$

$$= \sqrt{4} + \frac{2}{2\sqrt{4}}$$

$$= 2 + \frac{1}{2}$$

$$= \frac{5}{2}$$

$$\therefore \frac{dy}{dx} \text{ at } y = -2 \text{ is } \frac{2}{5} \quad \checkmark$$

(2 marks)

b) R.T.P. $\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}$ for $n \geq 1$
(where $n \in \mathbb{Z}$)

Prove true for $n=1$:

$$\text{LHS} = \frac{1}{2^1} = \frac{1}{2}$$

$$\text{RHS} = 2 - \frac{1+2}{2^1}$$

$$= 2 - \frac{3}{2}$$

$$= \frac{1}{2} = \text{LHS} \quad \therefore \text{true for } n=1$$

(1/2)

Assume true for $n=k$:

i.e. $\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k}{2^k} = 2 - \frac{k+2}{2^k}$

*

(1/2)

Prove true for $n=k+1$:

R.T.P. $\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}} = 2 - \frac{k+3}{2^{k+1}}$

L.H.S. $= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}}$ [using *]

$$= 2 - \left(\frac{k+2}{2^k} - \frac{k+1}{2^{k+1}} \right)$$

$$= 2 - \left[\frac{2(k+2) - (k+1)}{2^{k+1}} \right]$$

$$= 2 - \left(\frac{2k+4-k-1}{2^{k+1}} \right)$$

$$= 2 - \frac{k+3}{2^{k+1}}$$

$$= \text{RHS}$$

(3 marks)

$\therefore$ If true for $n=k$, then it is true for $n=k+1$.
Since it is true for $n=1$, then it is true for $n=2$
and so on for all integers $n \geq 1$ by Mathematical Induction.

c) when $y=0$: $\tan x - 1 = 0$

$$\tan x = 1$$

$$\therefore x\text{-int. is } \pi/4$$

$\frac{1}{2}$

$$V = \pi \int_0^{\pi/4} (\tan x - 1)^2 dx$$

$\frac{1}{2}$

$$V = \pi \int_0^{\pi/4} (\tan^2 x - 2 \tan x + 1) dx$$

$$V = \pi \int_0^{\pi/4} \left(\sec^2 x - \frac{2 \sin x}{\cos x} \right) dx$$

$$V = \pi \left[\tan x + 2 \ln |\cos x| \right]_0^{\pi/4}$$

✓

$$= \pi \left[\tan \frac{\pi}{4} + 2 \ln |\cos \frac{\pi}{4}| - (\tan 0 + 2 \ln |\cos 0|) \right]$$

$$= \pi \left[1 + 2 \ln \left(\frac{1}{\sqrt{2}} \right) \right] \text{ units}^3$$

✓

(3 marks)

$$= \pi \left[1 + 2 \ln 2^{-1/2} \right]$$

$$= \pi \left[1 + 2 \left(-\frac{1}{2} \right) \ln 2 \right]$$

$$= \pi \left[1 - \ln 2 \right] \text{ units}^3$$

$$d) \quad \frac{dV}{dt} = 12\pi$$

$$\text{If } V = \pi h(h+20)$$

$$V = \pi h^2 + 20\pi h$$

$$\text{then } \frac{dV}{dh} = 2\pi h + 20\pi$$

$$\text{Now } \boxed{\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}}$$

$$\frac{dh}{dt} = \frac{1}{2\pi h + 20\pi} \times 12\pi$$

$$\text{i.e. } \frac{dh}{dt} = \frac{12\pi}{2\pi(h+10)} = \frac{6}{h+10}$$

If the rate is $12\pi \text{ cm}^3/\text{s}$, then after 8 seconds, the volume increases by $8 \times 12\pi \text{ cm}^3$

$$\text{i.e. } V = 96\pi \text{ cm}^3$$

$$\text{solve for } h: \quad 96\pi = \pi h(h+20) \quad \left[\text{using } V = \pi h(h+20) \right]$$

$$96 = h^2 + 20h$$

$$h^2 + 20h - 96 = 0$$

$$(h+24)(h-4) = 0$$

$$h = -24 \text{ or } h = 4$$

$$\therefore \underline{h = 4 \text{ only } (h > 0)}$$

$$\begin{aligned} \text{sub. into } \frac{dh}{dt} : \quad \frac{dh}{dt} &= \frac{6}{4+10} \\ &= \frac{6}{14} \\ &= \underline{\underline{\frac{3}{7} \text{ cm/s}}} \end{aligned}$$

(3 marks)

e) * Equating horizontal components:

$$Vt \cos \theta = 12a$$

$$\therefore t = \frac{12a}{V \cos \theta} \quad \dots (1)$$

* Equating vertical components:

$$Vt \sin \theta - \frac{1}{2} g t^2 = \frac{12a}{5} \quad \dots (2)$$

sub. (1) $\rightarrow$ (2):

$$V \left(\frac{12a}{V \cos \theta} \right) \sin \theta - \frac{1}{2} g \left(\frac{12a}{V \cos \theta} \right)^2 = \frac{12a}{5}$$

$$12a \tan \theta - \frac{1}{2} g \left(\frac{144a^2}{V^2 \cos^2 \theta} \right) = \frac{12a}{5}$$

$$12a \tan \theta - \frac{72g a^2 \sec^2 \theta}{V^2} = \frac{12a}{5}$$

* Sub. $V = \sqrt{15ag}$:

$$12a \tan \theta - \frac{72a^2 g (1 + \tan^2 \theta)}{15ag} = \frac{12a}{5}$$

$$12 \tan \theta - \frac{24(1 + \tan^2 \theta)}{5} = \frac{12}{5}$$

$$60 \tan \theta - 24 - 24 \tan^2 \theta = 12$$

$$24 \tan^2 \theta - 60 \tan \theta + 36 = 0$$

$$2 \tan^2 \theta - 5 \tan \theta + 3 = 0$$

* Let $u = \tan \theta$: $2u^2 - 5u + 3 = 0$

$$(2u - 3)(u - 1) = 0$$

$$\therefore u = \frac{3}{2} \text{ or } u = 1$$

$$\tan \theta = \frac{3}{2} \text{ or } \tan \theta = 1$$

$$\therefore \theta = 56^\circ \text{ or } 45^\circ$$

(4 marks)

Question 14

$$a) \quad \text{If } x = (\sin \theta)^2 \quad \begin{cases} \text{when } x=0 \\ \theta=0 \end{cases}$$
$$\frac{dx}{d\theta} = 2 \sin \theta \cos \theta \quad \begin{cases} \text{when } x=1 \\ \theta=\frac{\pi}{2} \end{cases}$$

$$\therefore dx = 2 \sin \theta \cos \theta d\theta \quad \frac{1}{2}$$

$$\int_0^1 \sqrt{\frac{x}{1-x}} dx$$
$$= \int_0^{\pi/2} \left(\sqrt{\frac{\sin^2 \theta}{1-\sin^2 \theta}} \times 2 \sin \theta \cos \theta \right) d\theta \quad \frac{1}{\sqrt{2}}$$

$$= \int_0^{\pi/2} \left(\frac{\sin \theta}{\cos \theta} \times 2 \sin \theta \cos \theta \right) d\theta$$

$$= \int_0^{\pi/2} 2 \sin^2 \theta d\theta \quad \frac{1}{2}$$

$$= \int_0^{\pi/2} 2 \times \frac{1}{2} (1 - \cos 2\theta) d\theta \quad \frac{1}{2}$$

$$= \int_0^{\pi/2} (1 - \cos 2\theta) d\theta$$

$$= \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/2} \quad \checkmark$$

$$= \frac{\pi}{2} - \frac{\sin(2 \times \frac{\pi}{2})}{2} - \left(0 - \frac{\sin 0}{2} \right)$$

$$= \frac{\pi}{2} - \frac{0}{2} - 0 + 0$$

$$= \frac{\pi}{2} \quad \checkmark$$

(4 marks)

$$b) \tan^{-1}\left(\frac{1}{x}\right) + \tan^{-1}\left(\frac{1}{x+1}\right) = \frac{\pi}{4}$$

$$\text{Let } A = \tan^{-1}\left(\frac{1}{x}\right) \text{ and } B = \tan^{-1}\left(\frac{1}{x+1}\right)$$

$$\text{then } \tan A = \frac{1}{x} \text{ and } \tan B = \frac{1}{x+1}$$

$$\therefore \text{ solving } A + B = \frac{\pi}{4}$$

$$\Rightarrow \tan(A+B) = \tan \frac{\pi}{4}$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\frac{\frac{1}{x} + \frac{1}{x+1}}{1 - \frac{1}{x} \times \frac{1}{x+1}} = 1$$

$$\frac{x+1+x}{x(x+1)} = 1 - \frac{1}{x(x+1)}$$

$$\frac{2x+1}{x(x+1)} = \frac{x(x+1)-1}{x(x+1)}$$

$$2x+1 = x^2+x-1$$

$$x^2+x-2x-1-1=0$$

$$x^2-x-2=0$$

$$(x-2)(x+1)=0$$

$$\therefore x=2 \text{ or } x=-1$$

but $x=2$ is only valid solution

[since $\tan^{-1}\left(\frac{1}{-1+1}\right)$ is undefined]

(1/2)

(1/2)

(3 marks)

$$c) i) \frac{dP}{dt} = P e^{-0.5t}$$

$$\frac{1}{P} dP = e^{-0.5t} dt$$

$$\int \frac{1}{P} dP = \int e^{-0.5t} dt$$

$$\ln |P| = \frac{e^{-0.5t}}{-0.5} + C$$

$$\ln |P| = -2e^{-0.5t} + C$$

$$|P| = e^{-2e^{-0.5t} + C}$$

$$P = \pm e^C \times e^{-2e^{-0.5t}}$$

$$\therefore P = A e^{-2e^{-0.5t}}$$

$$[\text{where } A = \pm e^C]$$

$$(ii) \text{ As } t \rightarrow \infty$$

$$P \rightarrow A e^{-2 \times 0}$$

$$\therefore P \rightarrow A$$

$$\text{Now find } t \text{ when } P = \frac{1}{2} A:$$

$$\frac{1}{2} A = A e^{-2e^{-0.5t}}$$

$$0.5 = e^{-2e^{-0.5t}}$$

$$\ln(0.5) = -2e^{-0.5t}$$

$$\ln(2)^{-1} = -2e^{-0.5t}$$

$$\ln 2 = 2e^{-0.5t}$$

$$\ln \left[\frac{\ln 2}{2} \right] = -0.5t$$

$$t = \ln \left[\frac{\ln 2}{2} \right] \div -0.5$$

$$\therefore t = -2 \ln \left[\frac{\ln 2}{2} \right] \quad (\text{ok. here})$$

$$\text{or } t = 2 \ln \left[\frac{2}{\ln 2} \right]$$

(2 marks)

$$\text{or } t = -2 \ln(\ln \sqrt{2})$$

(2 marks)

d) i) Since $AD:DC = 2:1$

$$\text{then } \vec{AD} = \frac{2}{3} \vec{c}$$

$$\text{and } \vec{DC} = \frac{1}{3} \vec{c}$$

$$\left\{ \begin{array}{l} \text{Since } BE:EC = 1:2 \\ \text{then } \vec{BE} = \frac{1}{3} \vec{BC} \\ \text{and } \vec{EC} = \frac{2}{3} \vec{BC} \end{array} \right.$$

$$\begin{aligned} \text{Now } \vec{BC} &= \vec{BA} + \vec{AC} \\ &= -\vec{b} + \vec{c} \end{aligned}$$

$$\therefore \vec{BC} = \vec{c} - \vec{b}$$

*

$$\text{And } \vec{DE} = \vec{DC} + \vec{CE}$$

$$\vec{DE} = \vec{DC} - \vec{EC}$$

$$\vec{DE} = \vec{DC} - \frac{2}{3} \vec{BC}$$

$$\vec{DE} = \frac{1}{3} \vec{c} - \frac{2}{3} (\vec{c} - \vec{b})$$

$$\vec{DE} = \frac{1}{3} \vec{c} - \frac{2}{3} \vec{c} + \frac{2}{3} \vec{b}$$

$$\therefore \vec{DE} = \frac{2}{3} \vec{b} - \frac{1}{3} \vec{c} \quad (\text{as required})$$

$\frac{1}{2}$

using *

$\frac{1}{2}$

(1 mark)

(ii) Method (1)

or

$$\vec{DF} = k \vec{DE}$$

$$= k \left(\frac{2}{3} \underline{\vec{b}} - \frac{1}{3} \underline{\vec{c}} \right)$$

$$\vec{AF} = \lambda \vec{AB}$$

$$= \lambda \underline{\vec{b}}$$

$$\text{Now } \vec{DF} = \vec{DA} + \vec{AF}$$

$$k \left(\frac{2}{3} \underline{\vec{b}} - \frac{1}{3} \underline{\vec{c}} \right) = -\frac{2}{3} \underline{\vec{c}} + \lambda \underline{\vec{b}}$$

Equating coefficients:

$$-\frac{k}{3} = -\frac{2}{3} \quad (\text{for vector } \underline{\vec{c}})$$

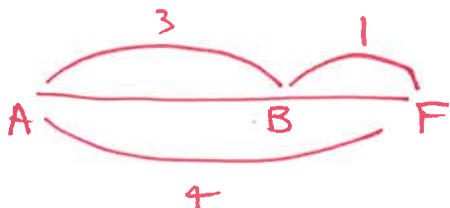
$$\therefore k = 2$$

and

$$\frac{2}{3} k = \lambda \quad (\text{for vector } \underline{\vec{b}})$$

$$\therefore \lambda = \frac{2}{3} \times 2$$

$$\lambda = \frac{4}{3}$$



$$\therefore AB : BF$$

$$= 3 : 1$$

(as required) (3 marks)

Method (2)

Since DEF is collinear and

$$\vec{DE} = \frac{2}{3} \underline{\vec{b}} - \frac{1}{3} \underline{\vec{c}}$$

$$\text{then } \vec{EF} = \lambda \left(\frac{2}{3} \underline{\vec{b}} - \frac{1}{3} \underline{\vec{c}} \right)$$

since ABF is collinear and $\vec{AB} = \underline{\vec{b}}$

$$\text{then } \vec{BF} = k \underline{\vec{b}}$$

Also, since $BE : EC = 1 : 2$,

$$\text{then } \vec{BE} = \frac{1}{3} \vec{BC}$$

$$\text{i.e. } \vec{BE} = \frac{1}{3} (\underline{\vec{c}} - \underline{\vec{b}})$$

$$\text{Now } \vec{BF} = \vec{BE} + \vec{EF}$$

$$\vec{BF} = \frac{1}{3} (\underline{\vec{c}} - \underline{\vec{b}}) + \lambda \left(\frac{2}{3} \underline{\vec{b}} - \frac{1}{3} \underline{\vec{c}} \right)$$

$$= \frac{1}{3} \underline{\vec{c}} - \frac{1}{3} \underline{\vec{b}} + \frac{2\lambda}{3} \underline{\vec{b}} - \frac{\lambda}{3} \underline{\vec{c}}$$

$$= \left(\frac{1-\lambda}{3} \right) \underline{\vec{c}} + \left(\frac{2\lambda-1}{3} \right) \underline{\vec{b}}$$

$$\text{but } \vec{BF} = k \underline{\vec{b}} + 0 \underline{\vec{c}}$$

$\therefore$ equating scalar coefficients:

$$\frac{1-\lambda}{3} = 0 \quad (\text{of } \underline{\vec{c}})$$

$$\therefore \lambda = 1$$

$$\text{and } \frac{2\lambda-1}{3} = k \quad (\text{of } \underline{\vec{b}})$$

$$\frac{2(1)-1}{3} = k$$

$$\therefore \underline{k = \frac{1}{3}}$$

$$\therefore \vec{BF} = \frac{1}{3} \underline{\vec{b}}$$

$$\therefore \vec{AB} : \vec{BF} = \underline{\vec{b}} : \frac{1}{3} \underline{\vec{b}}$$

$$= 3 : 1 \quad (\text{as required}) \quad (3 \text{ marks})$$